3D WEBSOFT INC. FOUNDATION

SCHOOL FUTURE TECH PROGRAM

Faculty Name :

Course Duration : 5 Months (150 Days)

Classes : Offline & Online

Daily Teaching : 3 Hours

Total Hours : 300 Hours

LEVEL 3 –Future Engineers(9th–10th Standard)

MODULE 1 – Python Foundations (Classes 1–10)

Class 5 – Python Syntax , Keywords & Commands

Complete Theory & Practical Notes

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Learning Objectives

By the end of this session, the class aims to:

1. Introduce students to the basic rules of Python syntax.
2. Help students understand how Python code is structured.
3. Explain the importance of correct spelling, symbols, spaces, and punctuation in Python.
4. Introduce Python keywords and explain their purpose.
5. Help students identify commonly used Python keywords.
6. Introduce single-line and multi-line comments.
7. Explain why programmers use comments in their programs.
8. Develop students' ability to read and identify basic Python syntax.
9. Provide hands-on practice in writing syntactically correct Python code.
10. Develop basic debugging skills by identifying common syntax mistakes.

Session

Topic	*Type*
Warm-up & Previous Knowledge Check	Discussion
Understanding Python Syntax	Theory
Python Syntax Rules	Theory + Demonstration
Common Syntax Errors	Demonstration
Practical 1 – Syntax Practice	Hands-on
Break / Syntax Challenge	Activity
Python Keywords	Theory + Demonstration
Practical 2 – Keyword Identification	Hands-on
Python Comments	Theory + Practical
Quiz + Exit Activity	Assessment

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1. Warm-up & Previous Knowledge Check

Start by asking students:

- What is Python?
- What is a Python program?
- What is a `.py` file?
- What does `print()` do?
- What happens when we make a mistake in our Python code?

```
print("Hello Python")
```

“What would happen if I remove one quotation mark?”

Then demonstrate:

```
print("Hello Python)
```

Computers need instructions to follow a specific set of rules. These rules are called syntax.

2. Understanding Python Syntax

What is Syntax?

Syntax is the set of rules that determine how Python code must be written.

Just like a human language has grammar rules, Python has programming rules.

Human Language

Incorrect:

“Python learning am I.”

Correct:

“I am learning Python.”

Python

Correct:

```
print("Hello")
```

Incorrect:

```
print("Hello"
```

The second example is missing a closing parenthesis.

Simple Analogy

Explain to students:

Syntax in programming is like grammar in a language.

If grammar is incorrect, a sentence may not make sense.

If Python syntax is incorrect, Python may not be able to execute the program.

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3. Basic Python Syntax Rules

THEORY + DEMONSTRATION

Introduce students to the most important beginner syntax rules.

Rule 1 – Correct Spelling

Correct:

```
print("Hello")
```

Incorrect:

```
pritn("Hello")
```

Python does not understand `pritn()`.

Rule 2 – Parentheses

Correct:

```
print("Hello")
```

Incorrect:

```
print("Hello"
```

Explain that `(` must normally have a matching `)`.

Rule 3 – Quotation Marks

Correct:

```
print("Python")
```

Incorrect:

```
print(Python)
```

At this stage, explain simply:

Text is normally placed inside quotation marks.

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Rule 4 – Matching Quotes

Correct:

```
print("Hello Future Engineers")
```

Incorrect:

```
print("Hello Future Engineers)
```

Rule 5 – Python is Case-Sensitive

These are not necessarily the same:

```
print()
Print()
PRINT()
```

The built-in function is:

```
print()
```

Rule 6 – One Instruction Per Line

Example:

```
print("Hello")
print("Welcome")
print("Python is fun")
```

Explain that Python normally executes these instructions from top to bottom.

4. Common Syntax Errors

DEMONSTRATION

Tell students:

“Making mistakes is a normal part of programming.”

Show common mistakes.

Error 1 – Missing Parenthesis

```
print("Hello"
```

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Correct:

```
print("Hello")
```

Error 2 – Missing Quotation Mark

```
print("Hello)
```

Correct:

```
print("Hello")
```

Error 3 – Misspelled Function

```
pritn("Hello")
```

Correct:

```
print("Hello")
```

Error 4 – Incorrect Capitalization

```
Print("Hello")
```

Correct:

```
print("Hello")
```

Error 5 – Unclosed String

```
print("Python is fun)
```

Correct:

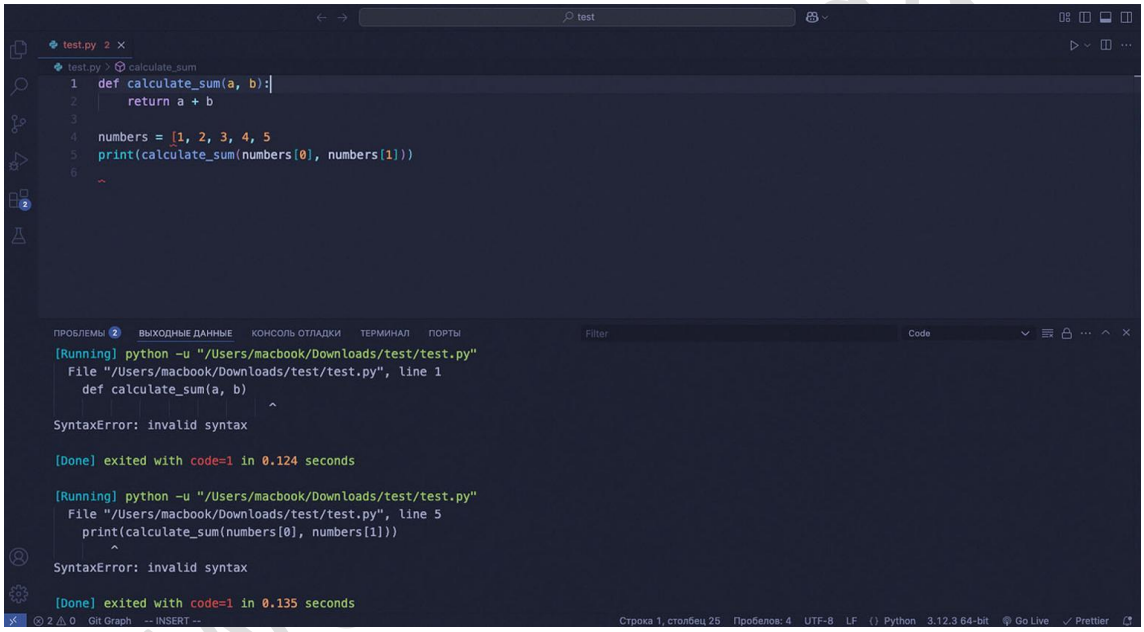
```
print("Python is fun")
```

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Suitable Visuals – Python Syntax



```
name = "Alice"
age = 25
print("Name:", name)
print("Age:", age)
```



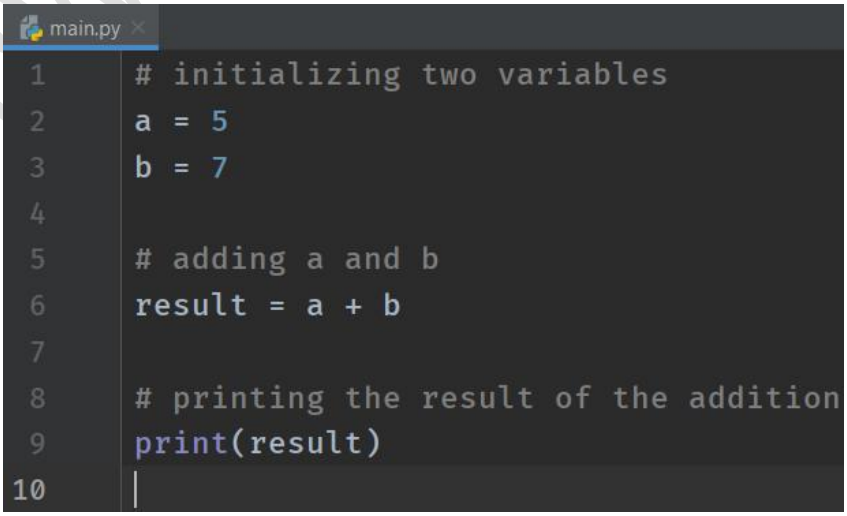
```
test.py 2
test.py > calculate_sum
1  def calculate_sum(a, b):
2      return a + b
3
4  numbers = [1, 2, 3, 4, 5
5  print(calculate_sum(numbers[0], numbers[1]))
6      ^

ПРОБЛЕМЫ 2  ВЫХОДНЫЕ ДАННЫЕ  КОНСОЛЬ ОТЛАДКИ  ТЕРМИНАЛ  ПОРТЫ
[Running] python -u "/Users/macbook/Downloads/test/test.py"
  File "/Users/macbook/Downloads/test/test.py", line 1
    def calculate_sum(a, b)
                           ^
SyntaxError: invalid syntax

[Done] exited with code=1 in 0.124 seconds

[Running] python -u "/Users/macbook/Downloads/test/test.py"
  File "/Users/macbook/Downloads/test/test.py", line 5
    print(calculate_sum(numbers[0], numbers[1]))
        ^
SyntaxError: invalid syntax

[Done] exited with code=1 in 0.135 seconds
```



```
main.py
1   # initializing two variables
2   a = 5
3   b = 7
4
5   # adding a and b
6   result = a + b
7
8   # printing the result of the addition
9   print(result)
10
```

5. Practical 1 – Syntax Practice

Students open VS Code and create

`syntax_practice.py`

Ask them to type:

```
print("Welcome to Python")
print("I am learning programming")
print("Python is exciting")
print("I want to become a programmer")
```

Students run the program.

Expected Output

```
Welcome to Python
I am learning programming
Python is exciting
I want to become a programmer
```

Student Task

Students intentionally make these mistakes one at a time:

1. Remove a quotation mark.
2. Remove a parenthesis.
3. Change `print` to `pritn`.
4. Change `print` to `Print`.

Then ask students to fix each error.

Teacher Observation

Check whether students can:

Recognize syntax errors
Read the error message
Locate the problematic line
Correct spelling
Correct quotation marks
Correct parentheses
Run the program again

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6. Break / Syntax Challenge

Give students a short challenge.

Display:

```
print("My name is Rahul)
print("I love Python"
pritn("I am a Future Engineer")
Print("Coding is fun")
```

Students work individually or in pairs.

Answer

There are four main errors:

- Missing quotation mark
- Missing closing parenthesis
- `pritn` instead of `print`
- `Print` instead of `print`

7. Python Keywords

THEORY + DEMONSTRATION

What is a Keyword?

A **keyword** is a reserved word in Python that has a special meaning to the language.

Explain using a simple analogy:

“Just like traffic signs have special meanings, Python keywords have special meanings inside Python programs.”

Examples include:

```
if
else
for
while
in
def
return
class
import
True
False
None
```

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Important Point

Students should not use Python keywords as ordinary variable names.

For example, keywords such as:

```
if
for
class
return
```

have predefined meanings in Python.

Beginner-Friendly Keyword Table

Keyword	Simple Meaning
`if`	Makes a decision
`Else`	Alternative condition
`for`	Repeats through items
`while`	Repeats while a condition is true
`in`	Checks membership
`def`	Defines a function
`return`	Sends a result back
`class`	Defines a class
`import`	Brings in a module
`True`	Boolean true value
`False`	Boolean false value
`None`	Represents no value

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8. Practical 2 – Keyword Identification

Give students this code:

```
 if age >= 18:
    print("Adult")

else:
    print("Minor")
```

Ask students to identify the keywords.

Expected Answers

```
if
else
```

Next:

```
for number in numbers:
    print(number)
```

Ask:

“Which words are Python keywords?”

Expected:

```
for
in
```

Classroom Game – “Keyword or Not?”

Display words one at a time:

Word	Keyword?
`if`	Yes
`hello`	No
`for`	Yes
`Python`	No
`class`	Yes
`student`	No

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Word	Keyword?
`return`	Yes
`computer`	No

Students respond by raising their hands or using cards marked **YES / NO**.

9. Python Comments

THEORY + PRACTICAL

What is a Comment?

A **comment** is a note written inside the program for humans to understand the code.

Python ignores comments when executing the program.

Single-Line Comment

Use `#`.

Example:

```
# This program introduces Python
print("Hello Python")
```

Explain:

```
# This is a comment
```

The comment is for the programmer.

Python does not execute the comment as an instruction.

Why Do We Use Comments?

Comments can help programmers:

- Explain what code does
- Remember important information
- Make programs easier to understand
- Help other programmers read the code
- Organize larger programs

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10. Practical – Add Comments to a Program

Students create:

```
comments_practice.py
```

and write:

```
# My first Python program

print("Hello, Future Engineer!")

# Display my learning goal
print("I am learning Python")

# Display my dream career
print("I want to become an engineer")
```

Ask students to add **three of their own comments**.

Example:

```
# My favorite subject
print("Science")

# My favorite hobby
print("Drawing")

# My future goal
print("I want to build robots")
```

11. Comment Challenge

Give students this code:

```
print("Welcome to Python")
print("My name is Alex")
print("I am learning coding")
print("I love technology")
```

Challenge:

Add a useful comment above every line.

Possible solution:

```
# Welcome message
print("Welcome to Python")

# Student name
print("My name is Alex")
```

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```
# Learning activity
print("I am learning coding")

# Interest
print("I love technology")
```

12. Final 5-Minute Assessment

Ask students to complete this independently:

```
# My Python Program

print("My name is __________")
print("I am learning Python")
print("Python is __________")
print("My dream is to become __________")
```

Question 1

What is Python syntax?

Question 2

What is a Python keyword?

Question 3

What symbol is used for a single-line comment?

Question 4

Find the error:

```
pritn("Hello")
```

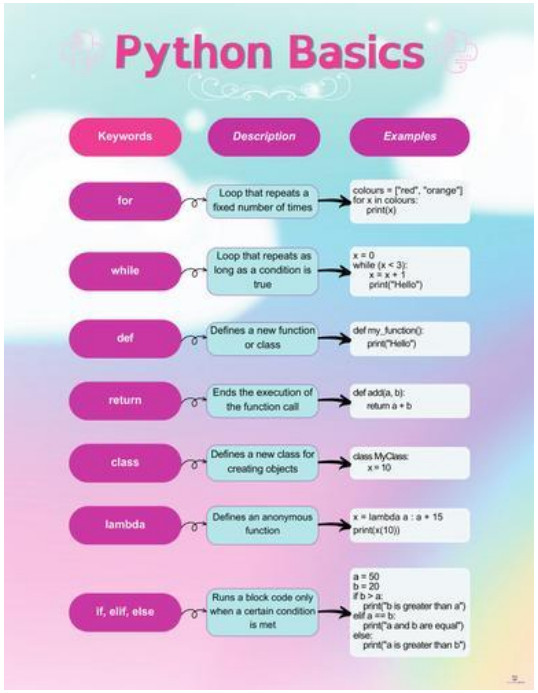
Question 5

Find the error:

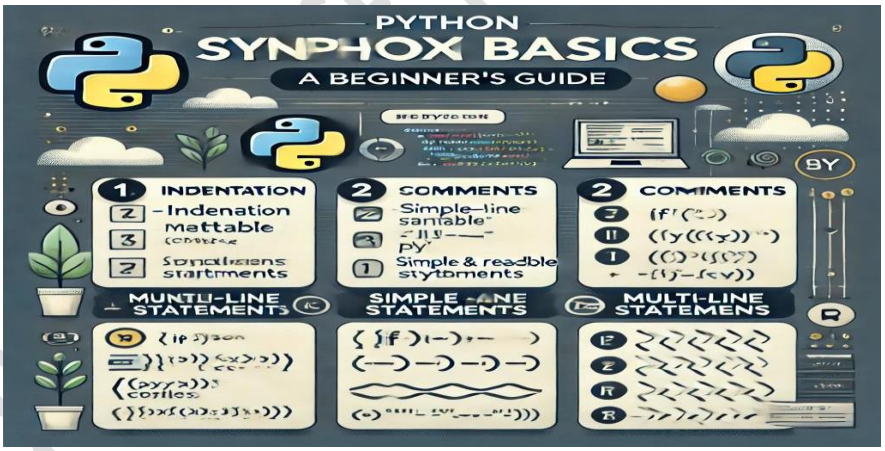
```
print("Hello)
```

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13. Suggested Images for the Lesson







Learning Outcomes

By the end of the class, students will be able to:

1. **Explain** what Python syntax means.
2. **Identify** basic rules used when writing Python code.
3. **Write** simple Python statements using correct syntax.
4. **Use** parentheses and quotation marks correctly.

5. **Recognize** that Python is case-sensitive.
6. **Identify** common Python syntax errors.
7. **Correct** basic errors involving spelling, quotation marks, and parentheses.
8. **Explain** what a Python keyword is.
9. **Recognize** commonly used Python keywords such as `if`, `else`, `for`, `in`, `def`, and `return`.
10. **Explain** the purpose of comments in a Python program.
11. **Create** single-line comments using `#`.
12. **Add meaningful comments** to their own Python programs.
13. **Read and understand** simple commented Python code.
14. **Demonstrate basic debugging skills** by identifying and correcting simple Python errors.
15. **Create and run** a small Python program using correct syntax and comments.

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